Debt Expansion Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Expansion Restraint (DER), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on DER achieves an annualized gross (net) Sharpe ratio of 0.48 (0.37), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (19) bps/month with a t-statistic of 3.22 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets) is 22 bps/month with a t-statistic of 3.01.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency of this process remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly incorporate all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This gradual diffusion manifests particularly in corporate financial decisions, where complex signals about firm fundamentals may take time to be fully understood and incorporated into prices.

We examine how firms' debt expansion restraint—measured as the negative change in long-term debt scaled by total assets—predicts future stock returns. Despite the fundamental importance of capital structure decisions for firm value, markets appear to systematically underreact to the information contained in debt financing choices. This underreaction creates a persistent return premium for firms exhibiting debt expansion restraint, suggesting that investors fail to immediately recognize the full implications of conservative debt management for future profitability and risk.

The slow diffusion of information provides a compelling framework for understanding why debt expansion restraint predicts returns. When firms restrain debt expansion, they signal multiple positive attributes: financial discipline, confidence in internal cash generation, and reduced financial risk. However, these signals require careful analysis of financial statements and may not be immediately salient to all investors. Following the limited attention framework of Hirshleifer and Teoh (2003) and DellaVigna and Pollet (2009), investors facing cognitive constraints prioritize easily processed information while neglecting more complex signals embedded in financing activities. This selective attention creates systematic underreaction to debt restraint signals, allowing informed investors to earn abnormal returns.

The gradual information diffusion mechanism predicts specific cross-sectional patterns in return predictability. Hong et al. (2000) demonstrate that information travels slowly from informed to uninformed investors, with the speed of diffusion depending on firm visibility and investor sophistication. For debt expansion restraint, this implies stronger return predictability among firms with lower analyst coverage, smaller market capitalizations, and higher information asymmetry. These firms face greater frictions in information transmission, as documented by Hou (2007), who show that information diffuses more slowly for neglected stocks. The complexity of interpreting debt financing decisions exacerbates these frictions, as investors must disentangle strategic motives from financial constraints.

The underreaction to debt restraint signals should generate return continuation rather than reversal patterns. Unlike behavioral overreaction stories that predict eventual price corrections, the gradual information diffusion hypothesis suggests that prices drift toward fundamental values as information spreads. Zhang (2006) provides a foundation for this mechanism, showing that information uncertainty amplifies underreaction to value-relevant signals. Debt expansion restraint represents precisely such a signal: it conveys information about future cash flows and discount rates, but this information only gradually becomes reflected in prices as more investors recognize its implications. The resulting return continuation should be most pronounced in segments of the market where information frictions are strongest.

Our empirical analysis reveals that debt expansion restraint strongly predicts cross-sectional stock returns. The value-weighted long-short portfolio that buys high DER stocks and sells low DER stocks earned an average return of 26 basis points per month with a t-statistic of 3.45 over the 1973-2024 sample period. This strategy achieved an annualized Sharpe ratio of 0.48, placing it in the top 13% of documented anomalies. The return premium remains robust after controlling for standard risk factors: the alpha relative to the Fama-French six-factor model equals 25 basis points per month with a t-statistic of 3.22. These results demonstrate economically significant compensation for holding stocks with high debt expansion restraint.

The predictive power of debt expansion restraint extends across different segments of the market and remains robust to alternative portfolio construction methods. Among the largest quintile of stocks by market capitalization, the DER strategy generated an average return of 34 basis points per month with a t-statistic of 3.65. The alpha for these large-cap stocks ranges from 25 to 36 basis points per month across standard factor models, with t-statistics exceeding 2.64. This performance among large, liquid stocks suggests that the DER premium cannot be attributed solely to market microstructure effects or limits to arbitrage. Even after accounting for transaction costs using the composite bid-ask spreads from Chen and Velikov (2022), the net Sharpe ratio of 0.37 ranks in the top 6% of anomalies.

Debt expansion restraint provides incremental predictive power beyond closely related anomalies. Controlling for six related financing and investment signals—including net debt financing, change in net operating assets, and inventory growth—the DER strategy maintains an alpha of 22 basis points per month with a t-statistic of 3.01. This distinctiveness extends to the broader factor zoo: when combined with 155 existing anomalies using the average rank method, adding DER increases terminal wealth by 25%. The signal's correlation with existing predictors remains modest, with the highest absolute correlation reaching only 0.31. These findings establish debt expansion restraint as a novel and economically important predictor that expands the investment opportunity set.

This paper contributes to three strands of the asset pricing literature. First, we extend the evidence on gradual information diffusion by identifying a new channel through which complex financial information slowly incorporates into prices. While Hong et al. (2000) focus on momentum in analyst forecasts and Cohen and Frazzini (2008) examine customer-supplier links, we show that debt financing decisions contain value-relevant information that diffuses gradually through markets. Our results complement Richardson et al. (2005) and Hirshleifer et al. (2004), who document market inefficiencies related to accounting information, by demonstrating that financing activities provide an additional source of return predictability arising from limited investor attention.

Methodologically, we advance the literature by conducting comprehensive robustness tests that address concerns about data mining and spurious predictability. Following the protocol of Novy-Marx and Velikov (2023), we evaluate debt expansion restraint across multiple portfolio construction methods, factor models, and transaction cost specifications. Our finding that DER maintains significant alpha after controlling for the six most closely related anomalies from the factor zoo addresses the multiple testing concerns raised by Harvey et al. (2016). The strategy's strong performance net of transaction costs, particularly among large-cap stocks, distinguishes it from many documented anomalies that exist primarily among illiquid securities.

Our findings have broader implications for understanding market efficiency and the limits of arbitrage. The persistence of the debt expansion restraint premium over five decades, despite the proliferation of quantitative investment strategies, suggests that some forms of accounting information remain systematically underexploited. This evidence supports models of limited attention such as Peng and Xiong (2006), where investors must allocate scarce cognitive resources across multiple information sources. For practitioners, our results identify an implementable investment strategy that generates significant abnormal returns even in the most liquid segment of the market. For researchers, the gradual incorporation of debt restraint information into prices provides a laboratory for testing theories about information processing and price formation in financial markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures.Debt Expansion Restraint signal relies on two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This item captures new long-term borrowing activities and excludes debt refinancing that does not generate net cash proceeds. Total assets (AT) represents the book value of all assets reported on the balance sheet at fiscal year-end, providing a comprehensive measure of firm size and serving as a natural scaling variable for financial statement items.construct the Debt Expansion Restraint signal by calculating the year-over-year change in long-term debt issuance, scaled by lagged total assets. Specifically, we compute (DLTIS(t) - DLTIS(t-1)) / AT(t-1) for each firm-year observation. This measure captures the acceleration or deceleration in a firm's debt financing activities relative to its asset base. A positive value indicates an increase in the pace of debt issuance compared to the previous year, while a negative value suggests restraint in debt expansion. We use annual observations based on fiscal year-end values and require non-missing values for both DLTIS and AT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with negative or zero lagged assets are excluded from the analysis.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DER signal. Panel A plots the time-series of the mean, median, and interquartile range for DER. On average, the cross-sectional mean (median) DER is -0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DER data. The signal's interquartile range spans -0.06 to 0.05. Panel B of Figure 1 plots the time-series of the coverage of the DER signal for the CRSP universe. On average, the DER signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does DER predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DER using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DER portfolio and sells the low DER portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DER strategy earns an average return of 0.26% per month with a t-statistic of 3.45. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.25% to 0.31% per month and have t-statistics exceeding 3.22 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 4.66 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 535 stocks and an average market capitalization of at least $1,410 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 4.67. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -5-20bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.82. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DER trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DER strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DER, as well as average returns and alphas for long/short trading DER strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the DER strategy achieves an average return of 34 bps/month with a t-statistic of 3.65. Among these large cap stocks, the alphas for the DER strategy relative to the five most common factor models range from 25 to 36 bps/month with t-statistics between 2.64 and 3.90.

5 How does DER perform relative to the zoo?

Figure 2 puts the performance of DER in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the DER strategy falls in the distribution. The DER strategy's gross (net) Sharpe ratio of 0.48 (0.37) is greater than 87% (94%) of anomaly Sharpe ratios,

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DER strategy (red line).[2] Ignoring trading costs, a \$1 invested in the DER strategy would have yielded \$3.21 which ranks the DER strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DER strategy would have yielded \$1.92 which ranks the DER strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DER relative to those. Panel A shows that the DER strategy gross alphas fall between the 58 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DER strategy has a positive net generalized alpha for five out of the five factor models. In these cases DER ranks between the 76 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does DER add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DER with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DER or at least to weaken the power DER has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DER} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DER} DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DER trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DER and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DER signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on DER is -1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the DER strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DER strategy earns alphas that range from 22-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DER trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.01.

7 Does DER add relative to the whole zoo?

Finally, we can ask how much adding DER to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DER signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes DER grows to $1243.77.

8 Conclusion

This study provides compelling evidence that Debt Expansion Restraint (DER) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that DER-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The strategy's annualized Sharpe ratio of 0.48 (gross) and 0.37 (net) indicates strong economic significance, while the persistence of alpha relative to the Fama-French five-factor model plus momentum (25 bps gross, 19 bps net) with robust t-statistics confirms that DER captures a distinct dimension of expected returns not explained by conventional risk factors. Particularly noteworthy is the signal's ability to generate significant alpha (22 bps with t-statistic of 3.01) even after controlling for six closely related financing and investment strategies, suggesting that DER provides unique information about firm fundamentals and future stock performance beyond existing measures of corporate financing and investment activities. From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magni-

ization on CRSP in the period for which DER is available.

tude and statistical robustness of the DER signal suggest it warrants consideration in asset pricing models and portfolio construction strategies. The relatively modest degradation from gross to net returns indicates that implementation costs, while non-trivial, do not eliminate the strategy's profitability, making it potentially viable for real-world application. However, several limitations merit acknowledgment. First, while our analysis controls for transaction costs, actual implementation may face additional frictions such as market impact and capacity constraints that could further erode returns. Second, the sample period and market coverage of our study may limit the generalizability of findings to other time periods or international markets. Third, the economic mechanisms underlying the DER anomaly remain incompletely understood, raising questions about whether the premium represents compensation for systematic risk or market inefficiency. Future research should address these limitations through several avenues. Investigation of the DER signal's performance in international markets would test its robustness across different institutional and regulatory environments. Time-varying analysis of the DER premium could reveal whether its predictive power varies with market conditions or macroeconomic cycles. Additionally, deeper exploration of the economic channels through which debt expansion restraint affects firm value and investor perceptions would enhance our theoretical understanding of this anomaly. Finally, examining the interaction between DER and corporate governance, financial constraints, or growth opportunities could provide insights into when and why the signal is most effective, potentially leading to refined trading strategies and better risk management frameworks.

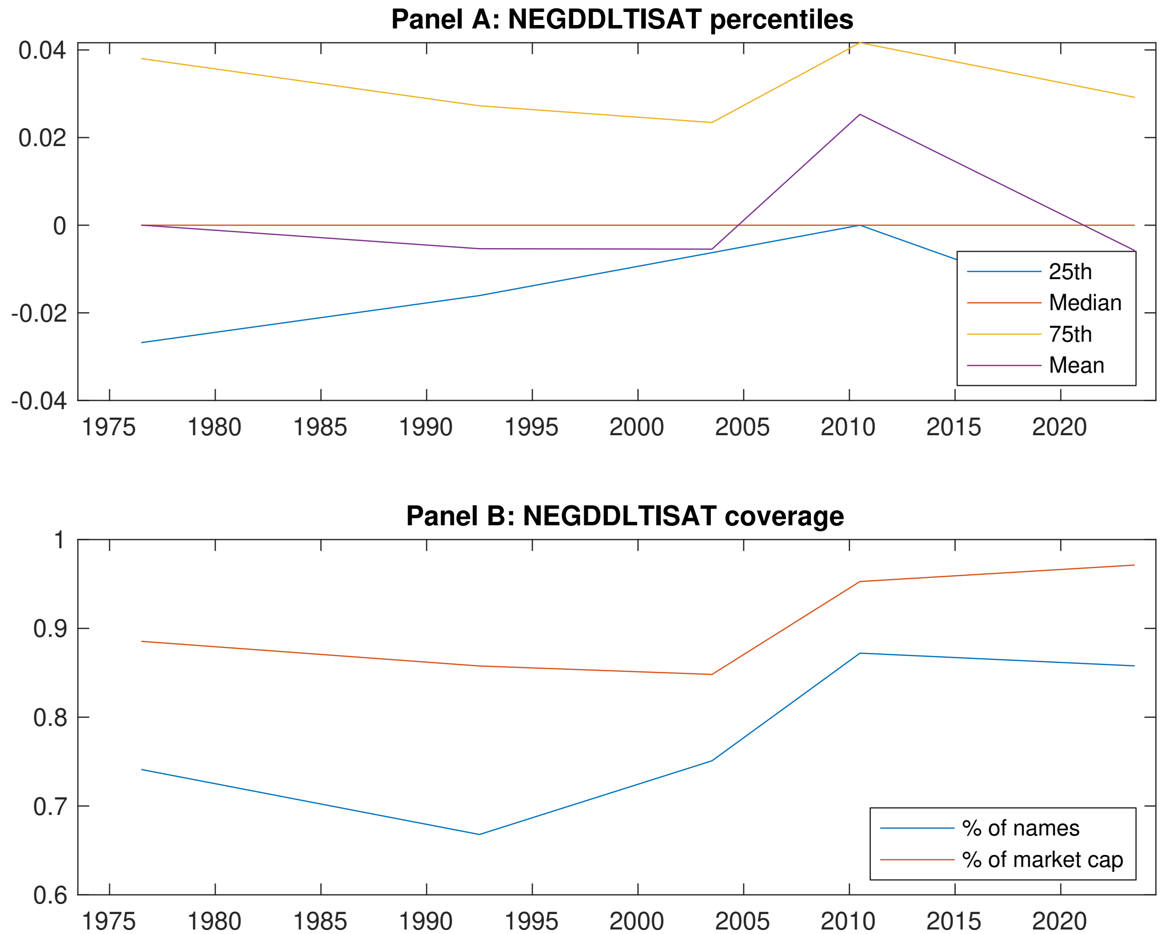


Figure 1: Times series of DER percentiles and coverage.
This figure plots descriptive statistics for DER. Panel A shows cross-sectional percentiles of DER over the sample. Panel B plots the monthly coverage of DER relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DER. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DER-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.59	0.64	0.62	0.72	0.85	0.26
	[2.81]	[3.53]	[3.13]	[4.04]	[4.26]	[3.45]
α_{CAPM}	-0.12	0.03	-0.05	0.11	0.18	0.30
	[-2.23]	[0.56]	[-0.89]	[2.49]	[3.11]	[4.02]
α_{FF3}	-0.12	-0.02	-0.00	0.09	0.19	0.31
	[-2.22]	[-0.40]	[-0.02]	[2.02]	[3.49]	[4.10]
α_{FF4}	-0.10	0.01	0.04	0.06	0.17	0.27
	[-1.92]	[0.12]	[0.82]	[1.31]	[3.05]	[3.58]
α_{FF5}	-0.12	-0.06	0.05	0.01	0.15	0.27
	[-2.25]	[-1.32]	[0.92]	[0.15]	[2.65]	[3.53]
α_{FF6}	-0.11	-0.04	0.08	-0.01	0.13	0.25
	[-2.05]	[-0.87]	[1.51]	[-0.26]	[2.41]	[3.22]
Panel B: Fama and French (2018) 6-factor model loadings for DER-sorted portfolios						
β_{MKT}	1.06	0.98	0.99	0.98	1.02	-0.04
	[83.62]	[96.26]	[77.85]	[97.21]	[79.29]	[-2.21]
β_{SMB}	0.16	-0.14	-0.02	-0.04	0.14	-0.01
	[8.39]	[-9.08]	[-0.87]	[-2.72]	[7.51]	[-0.55]
β_{HML}	-0.03	0.16	-0.10	0.02	-0.14	-0.11
	[-1.21]	[8.06]	[-4.23]	[1.22]	[-5.53]	[-3.16]
β_{RMW}	0.08	0.12	-0.05	0.11	0.02	-0.06
	[3.16]	[5.75]	[-2.01]	[5.53]	[0.82]	[-1.66]
β_{CMA}	-0.09	0.00	-0.11	0.16	0.15	0.24
	[-2.35]	[0.02]	[-2.90]	[5.38]	[4.09]	[4.66]
β_{UMD}	-0.02	-0.03	-0.05	0.03	0.02	0.03
	[-1.24]	[-3.09]	[-4.05]	[2.78]	[1.46]	[1.95]
Panel C: Average number of firms (n) and market capitalization (me)						
n	671	535	1060	590	641	
me ($\$10^6$)	1410	3195	2218	3114	1416	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the DER strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26	0.30	0.31	0.27	0.27	0.25
			[3.45]	[4.02]	[4.10]	[3.58]	[3.53]	[3.22]
Quintile	NYSE	EW	0.21	0.24	0.22	0.20	0.21	0.20
			[4.67]	[5.27]	[4.91]	[4.46]	[4.59]	[4.32]
Quintile	Name	VW	0.25	0.29	0.30	0.26	0.25	0.22
			[3.42]	[4.11]	[4.23]	[3.59]	[3.42]	[3.04]
Quintile	Cap	VW	0.21	0.25	0.25	0.20	0.17	0.15
			[3.29]	[3.82]	[3.78]	[3.10]	[2.70]	[2.29]
Decile	NYSE	VW	0.23	0.31	0.28	0.24	0.12	0.11
			[2.42]	[3.30]	[2.94]	[2.49]	[1.33]	[1.19]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.20	0.23	0.24	0.22	0.21	0.19
			[2.64]	[3.13]	[3.17]	[2.91]	[2.76]	[2.54]
Quintile	NYSE	EW	-0.05					
			[-0.82]					
Quintile	Name	VW	0.19	0.24	0.24	0.22	0.20	0.18
			[2.58]	[3.31]	[3.38]	[3.06]	[2.80]	[2.55]
Quintile	Cap	VW	0.16	0.20	0.19	0.17	0.13	0.12
			[2.48]	[3.06]	[2.98]	[2.63]	[2.09]	[1.81]
Decile	NYSE	VW	0.16	0.24	0.21	0.19	0.08	0.07
			[1.66]	[2.50]	[2.19]	[1.97]	[0.90]	[0.75]

Table 3: Conditional sort on size and DER

This table presents results for conditional double sorts on size and DER. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DER. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DER and short stocks with low DER .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		DER Quintiles					DER Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.65	0.86	0.94	0.87	0.83	0.17	0.20	0.19	0.14	0.15	0.12
		[2.36]	[3.20]	[3.41]	[3.18]	[2.91]	[1.86]	[2.16]	[1.99]	[1.44]	[1.62]	[1.28]
	(2)	0.72	0.90	0.82	0.88	0.87	0.14	0.18	0.15	0.16	0.14	0.14
		[2.69]	[3.65]	[3.22]	[3.58]	[3.42]	[1.78]	[2.26]	[1.96]	[1.98]	[1.73]	[1.78]
	(3)	0.78	0.79	0.81	0.85	0.90	0.12	0.19	0.18	0.14	0.16	0.13
		[3.06]	[3.62]	[3.35]	[3.79]	[3.83]	[1.52]	[2.41]	[2.28]	[1.75]	[1.98]	[1.62]
	(4)	0.73	0.78	0.84	0.74	0.91	0.17	0.21	0.20	0.18	0.19	0.17
		[3.18]	[3.75]	[3.84]	[3.55]	[4.15]	[2.31]	[2.75]	[2.68]	[2.28]	[2.39]	[2.12]
	(5)	0.50	0.60	0.58	0.61	0.83	0.34	0.36	0.36	0.29	0.29	0.25
		[2.56]	[3.36]	[2.94]	[3.35]	[4.42]	[3.65]	[3.88]	[3.90]	[3.15]	[3.13]	[2.64]

Panel B: Portfolio average number of firms and market capitalization

		DER Quintiles					DER Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	392	393	393	393	389	37	34	32	33	35
	(2)	107	107	107	108	107	61	61	60	61	61
	(3)	76	77	76	76	77	107	110	105	107	108
	(4)	64	65	64	65	64	229	240	229	235	230
	(5)	59	59	59	59	59	1453	2220	1800	2187	1517

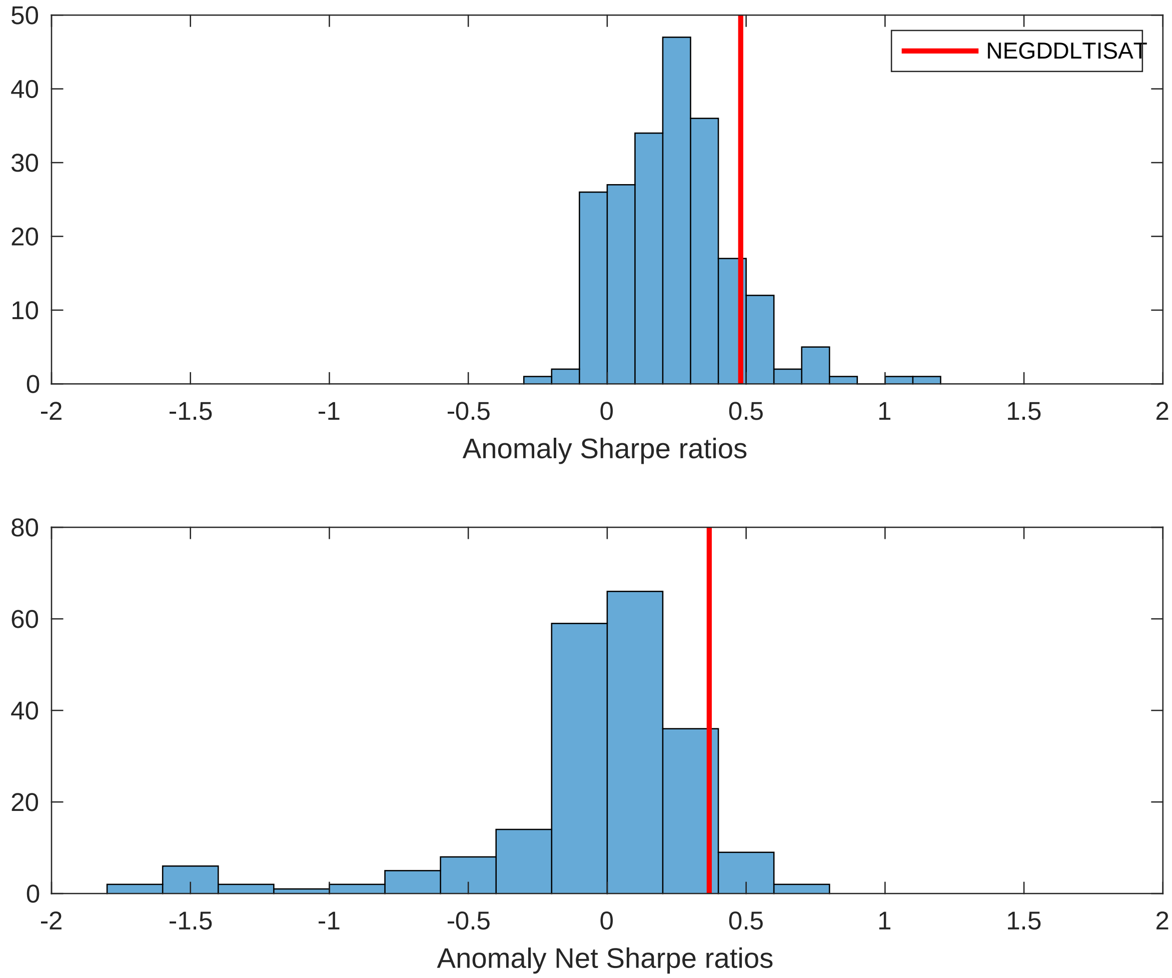


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DER with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

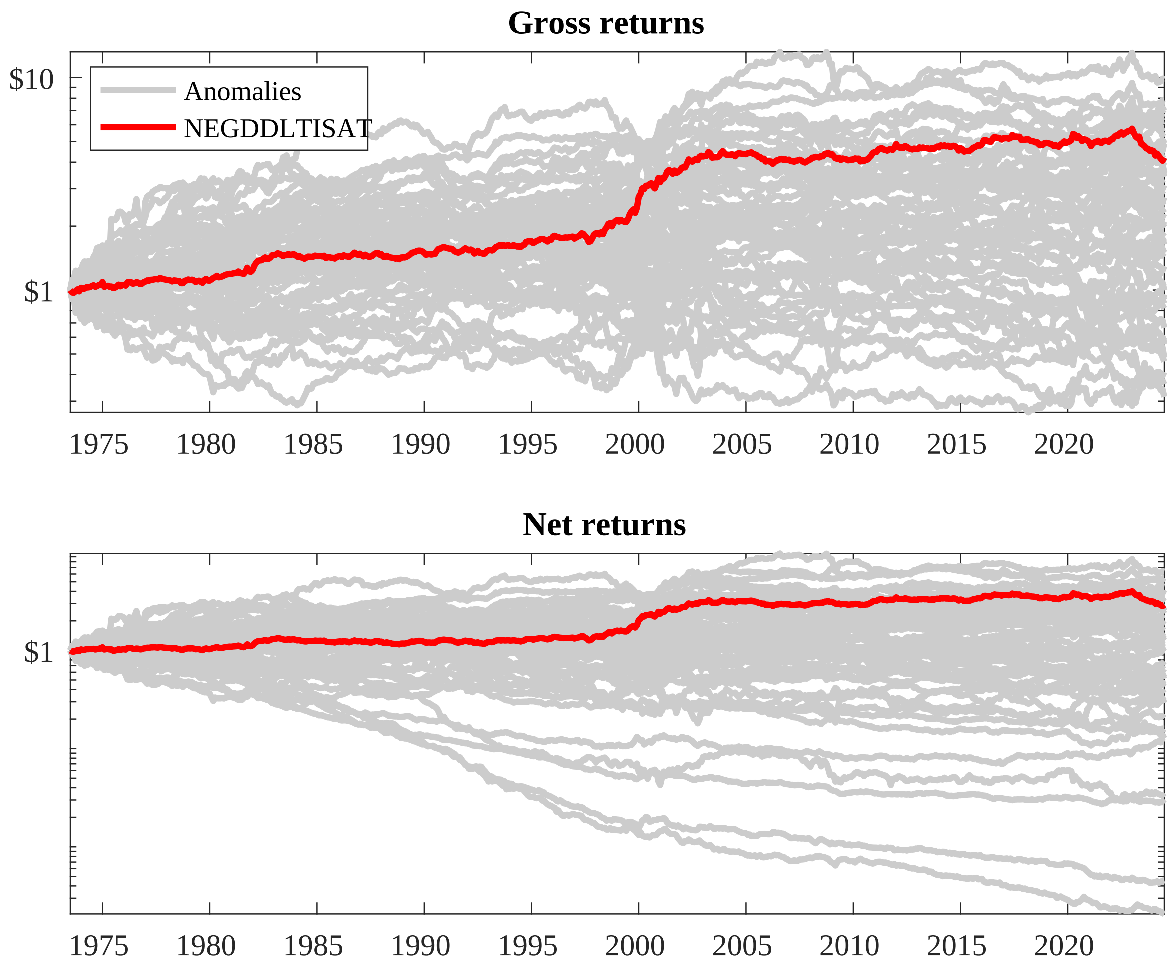
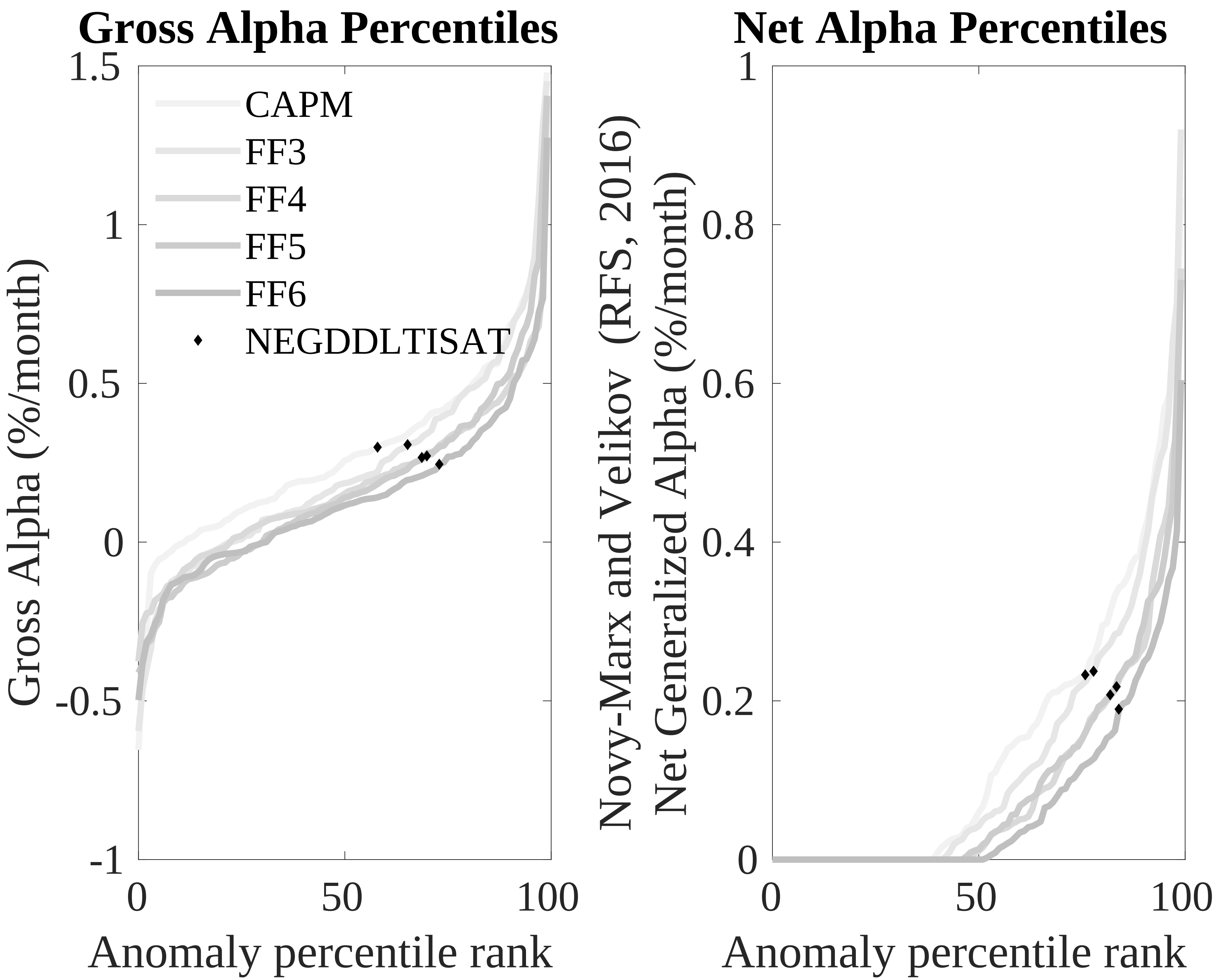


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DER trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISAT
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank

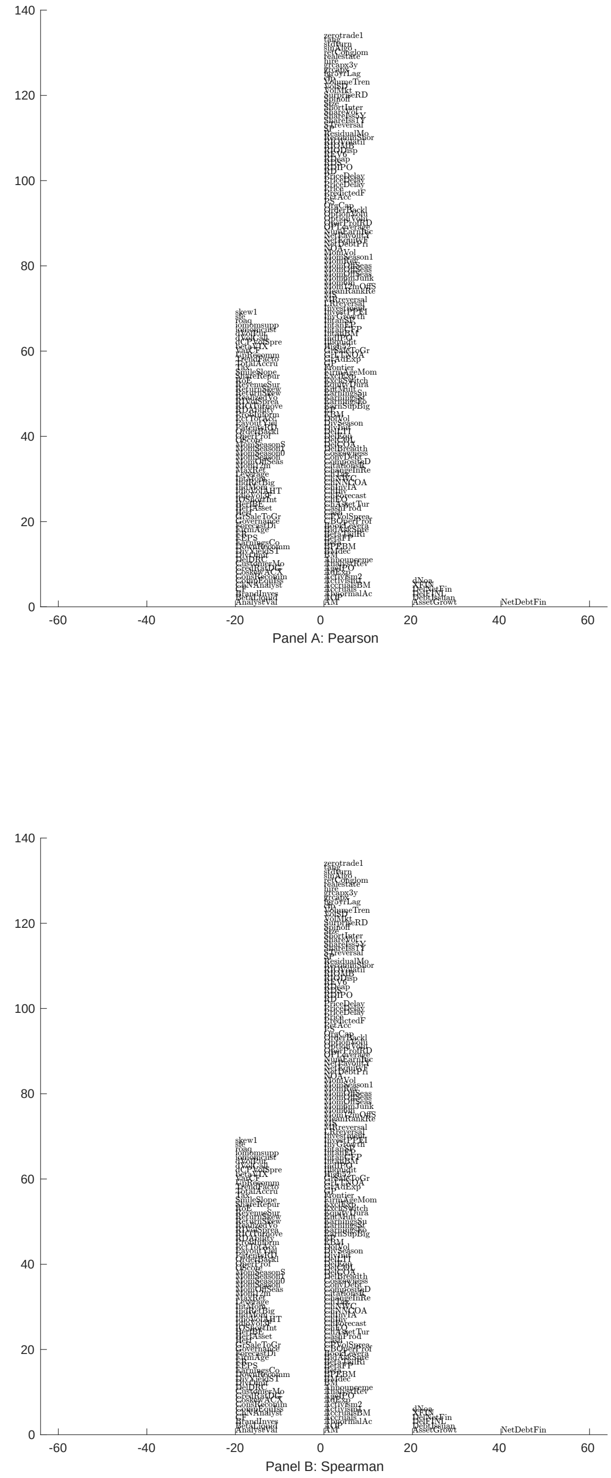


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with DER. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

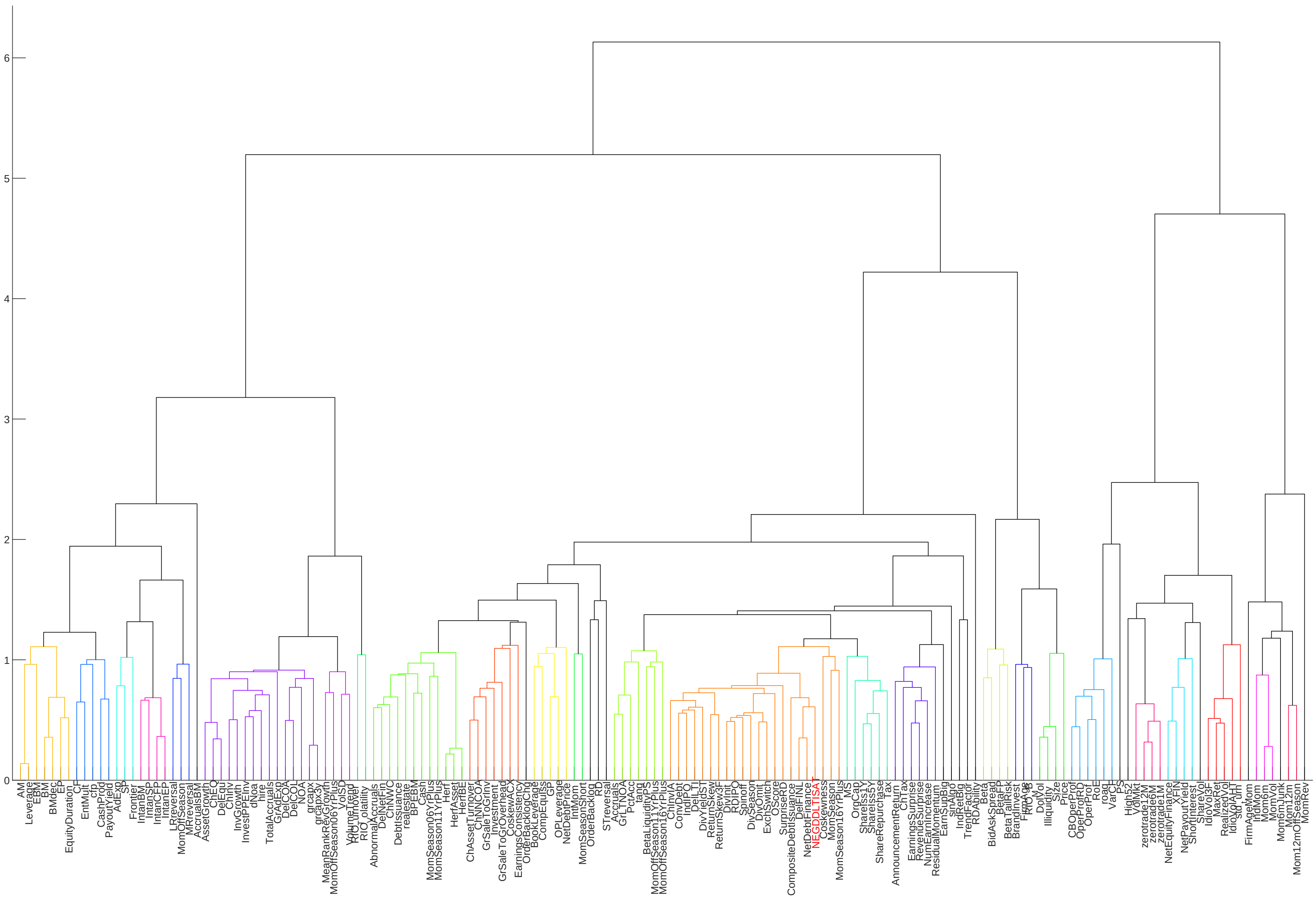


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

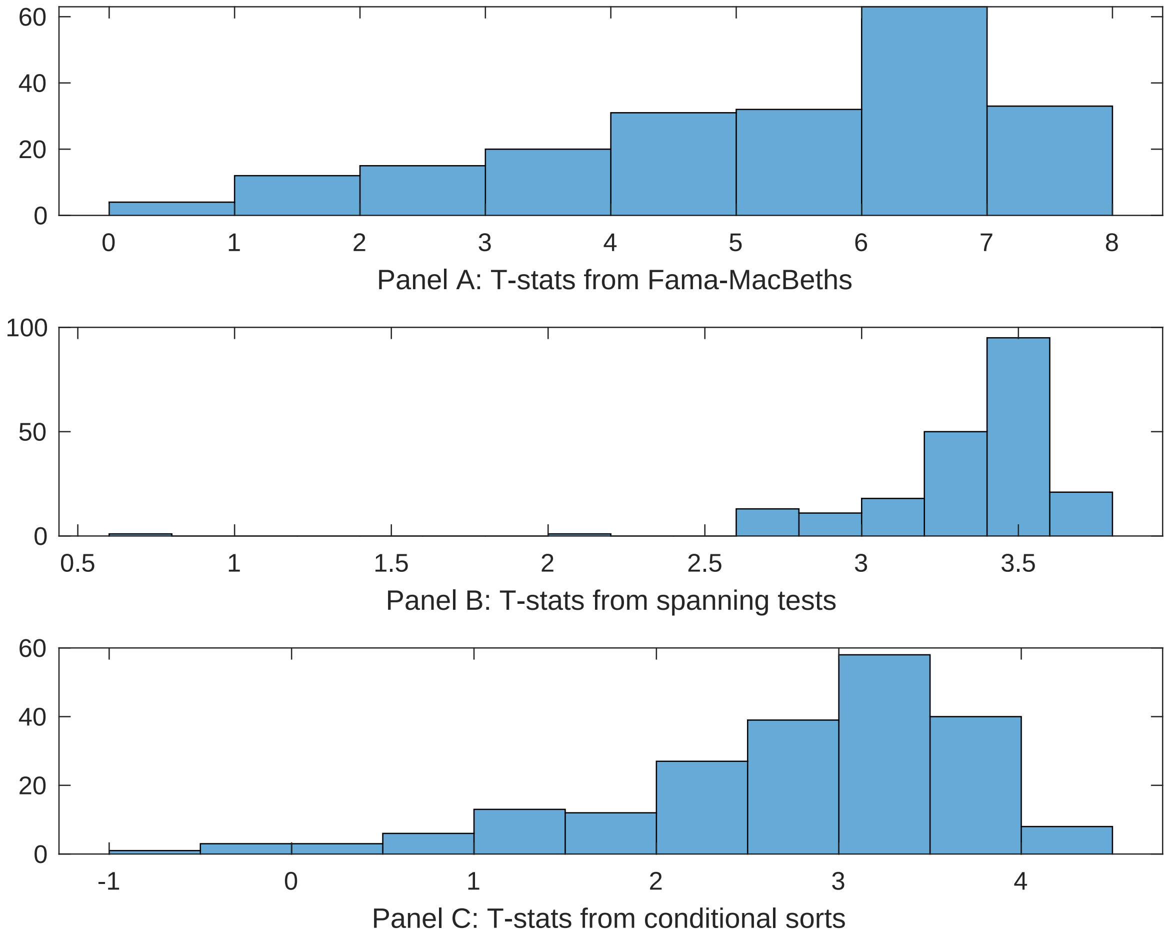


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DER} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DER} DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DER trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DER. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DER} DER_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.14	0.14	0.15
	[5.33]	[5.36]	[5.32]	[5.69]	[5.73]	[5.65]	[5.75]
DER	0.12	0.10	0.51	0.20	0.21	0.29	-0.14
	[1.21]	[1.11]	[5.11]	[1.88]	[2.33]	[0.31]	[-1.22]
Anomaly 1	0.20						0.11
	[8.54]						[1.59]
Anomaly 2		0.18					-0.10
		[9.14]					[-2.20]
Anomaly 3			0.40				0.21
			[6.87]				[0.35]
Anomaly 4				0.19			0.11
				[6.09]			[2.16]
Anomaly 5					0.17		0.69
					[7.67]		[2.29]
Anomaly 6						0.14	0.96
						[9.71]	[4.63]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DER trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DER} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.22	0.22	0.24	0.22	0.23	0.23	0.22
	[2.94]	[2.88]	[3.20]	[2.89]	[3.12]	[2.99]	[3.01]
Anomaly 1	19.46						10.38
	[4.77]						[1.87]
Anomaly 2		16.73					10.42
		[3.94]					[1.79]
Anomaly 3			10.13				9.19
			[3.46]				[2.99]
Anomaly 4				13.69			9.30
				[3.64]			[2.36]
Anomaly 5					9.28		4.22
					[2.72]		[1.12]
Anomaly 6						4.34	-8.15
						[1.00]	[-1.68]
mkt	-3.52	-3.17	-3.74	-1.68	-3.68	-3.54	-2.26
	[-2.05]	[-1.84]	[-2.16]	[-0.93]	[-2.12]	[-2.03]	[-1.27]
smb	-2.96	-2.96	-0.13	3.16	-1.21	-1.16	0.63
	[-1.13]	[-1.12]	[-0.05]	[1.10]	[-0.46]	[-0.44]	[0.22]
hml	-9.73	-9.41	-10.85	-9.00	-11.52	-10.88	-8.46
	[-2.94]	[-2.82]	[-3.25]	[-2.68]	[-3.42]	[-3.21]	[-2.52]
rmw	-7.67	-7.04	-4.06	-13.71	-5.36	-5.46	-11.89
	[-2.26]	[-2.07]	[-1.19]	[-3.37]	[-1.57]	[-1.60]	[-2.93]
cma	19.28	18.73	15.33	15.40	16.98	21.12	6.63
	[3.76]	[3.57]	[2.69]	[2.74]	[2.95]	[3.51]	[1.02]
umd	2.34	2.14	2.55	3.50	3.57	3.45	1.26
	[1.36]	[1.22]	[1.46]	[2.04]	[2.07]	[1.98]	[0.72]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	10	9	9	9	8	7	13

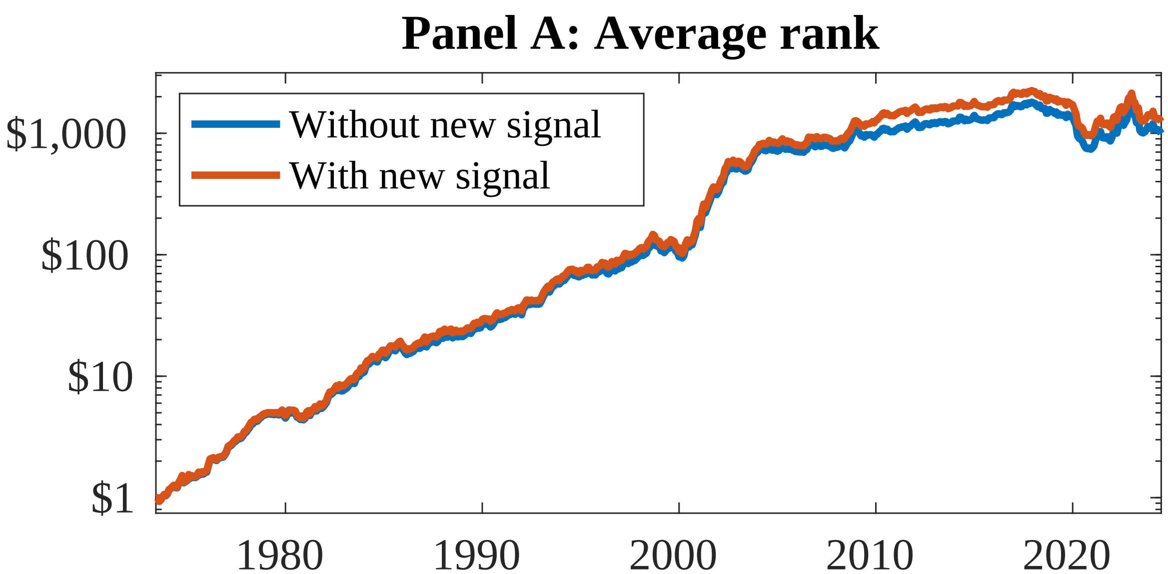


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DER. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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